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SiameseFWI: A Deep Learning Network for Enhanced Full Waveform Inversion

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Key Points:

- SiameseFWI, a deep learning approach, addresses the challenge of comparing simulated and observed seismic data in full-wave inversion (FWI) by transforming data into a shared latent space
- SiameseFWI introduces a self-supervised framework, leveraging the Siamese network, with a minor computational overhead compared to traditional FWI methods
- SiameseFWI demonstrates robust performance with synthetic and field data, highlighting its effectiveness in practical applications

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Abstract The performance of full-wave inversion (FWI) depends highly on how we compare the simulated data to observed ones. The simplified assumptions used to generate the simulated data make such comparison even harder. To address this challenge, we introduce SiameseFWI, a novel approach to FWI that plays a critical role in the comparative analysis of simulated and observed seismic data. Employing a Siamese network, this methodology transforms the data into a shared latent space, enabling a robust and effective comparison of data representations. SiameseFWI leverages two identical Convolutional Neural Networks with shared weights trained in a self-supervised framework, eliminating the necessity for labeled data. In each FWI iteration, the Siamese network and the velocity model are updated to minimize Euclidean distance loss between the latent representations of the data. Empirical evaluation conducted on the Marmousi2 and Overthrust models affirms the robust inversion performance of SiameseFWI compared to traditional FWI methodologies. Furthermore, its application to field data from Western Australia demonstrates its strength and efficacy in inversion. Notably, SiameseFWI exhibits robust inversion performance even in the presence of noise or when employing a linear initial model.

Plain Language Summary Full-wave inversion (FWI) aims to reveal subsurface structures by comparing the simulated and observed seismic data. However, this comparison is often complicated by the inherent differences between the naturally noise-prone field data and data simulated using our computational devices based on restrictive cost-effective assumptions (such as acoustic media). SiameseFWI, a novel approach, employs a Siamese network to transform data into a latent space for improved comparison. Rigorous evaluations of various data sets, including synthetic and field data, demonstrate its superior performance in inversion tasks. Importantly, despite SiameseFWI is self-supervised, it introduces minimal computational overhead compared to traditional FWI methods. This SiameseFWI innovation holds promise for enhancing our understanding of subsurface properties.

1. Introduction

Estimating the subsurface properties of the earth is crucial for various seismic applications including imaging and reservoir characterization. Full-wave inversion (FWI), a nonlinear inversion process, can provide detailed and high-resolution subsurface properties, which can be achieved through the iterative minimization of the difference between observed and simulated data (Tarantola, 1984; Virieux & Operto, 2009). In each iteration, FWI updates the velocity model so that the newly simulated data, through forward modeling, are closer to the observed ones. The efficacy of this update is influenced by the choice of the loss criterion in the FWI process, that is, a robust loss criterion yields superior FWI inversion results. Conventional FWI commonly uses the L2-norm misfit as its loss criterion, but this pixel-level comparison often leads the inversion to converge to a local minimum model. Various remedies have been proposed to address the cycle-skipping problem, including the introduction of customized loss criteria (Chi et al., 2014; Warner & Guasch, 2016) or the incorporation of regularization terms (Asnaashari et al., 2013; Esser et al., 2018; Guitton, 2012). For instance, employing the Wasserstein distance (Vallender, 1974) as a loss criterion has proven to enhance the robustness of FWI (Y. Yang et al., 2018).

Machine learning models have emerged as promising tools for serving as misfit functions in FWI. For example, a neural network was proposed for the misfit function (ML-misfit), employing a meta-learning procedure to train the neural network (Sun & Alkhalifah, 2022). In the meta-learning process, the ML-misfit network was trained through many FWI applications with diverse velocity models, where the true velocity models are known, and the L2-norm on the velocity fit is employed to optimize the ML-misfit model. As the ML-misfit is pre-trained, it can be smoothly integrated into FWI, measuring the loss without introducing a significant increase in the

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computational cost. However, to ensure the effectiveness of ML-misfit, a well-prepared training set is crucial to guarantee its ability to generalize across various data.

Furthermore, deep learning models serve as powerful tools for feature extraction from input data, with the convolutional neural network (CNN) standing out as one of the most extensively used models. CNN uses various learnable filters to extract multiple levels of abstractions, providing a well-represented depiction of the input data. This multi-level abstraction aids in capturing complex patterns and structures within the data, contributing to the effectiveness of the model in various applications. The CNN has been applied as a data-driven technique for predicting the 2D velocity model from the observed seismic data. During training, the CNN learns the nonlinearity mapping between the observed seismic data and the true velocity model (Kazei et al., 2021; Li et al., 2019; Liu et al., 2021; Wu & Lin, 2019; F. Yang & Ma, 2019). Similarly, inverting seismic data for the velocity model has been explored using a generative adversarial network (GAN) with transfer learning (Zhang & Lin, 2020). Additionally, the CNN has been proposed to be trained with synthetic data and updated during the FWI iterations (Wu et al., 2022). While some methods attempted to train the CNN using realistic velocity models, challenges arise due to the need for a large data set to ensure generalization and reliability. The application of CNN to field data in a supervised technique may encounter difficulties due to the absence of labels, and the observed data are often reasonably different from the synthetic training data (Sun & Alkhalifah, 2022).

Conversely, CNN and deep neural network have been employed with a deep image prior (DIP) concept, eliminating the need for labels, to predict the velocity model (He & Wang, 2021; Wu & McMechan, 2019). In this methodology, the weights of the deep learning models are initially trained by reconstructing the initial velocity model using noise as input data. Subsequently, the parameters of the pre-trained models are iteratively updated during the FWI process, aiming to reconstruct an improved velocity model that minimizes the misfit between observed and simulated data. Similarly, a generative neural network has been proposed to generate the velocity model and incorporate it into the partial differential equation solvers (Zhu et al., 2022). In this process, the weights of the model are updated by backpropagating gradients through the network layers and the acoustic wave equations. However, the DIP method requires a large number of FWI iterations to optimize the CNN parameters and obtain a reliable velocity model, adding a significant cost to the FWI process. Additionally, the DIP networks necessitate early stopping criteria to avoid producing unreliable outputs.

On the other hand, the Wasserstein Generative Adversarial Network with Gradient Penalty (WGAN-GP) employed a self-supervised approach to measure the loss for FWI (F. Yang & Ma, 2023). Unlike the ML-misfit, WGAN-GP does not require a training data set; it optimizes its parameters during the FWI process. WGAN-GP comprises a critic network, a combination of a CNN and fully connected layers, to measure differences between simulated and observed data. However, WGAN-GP adds significant computational cost to FWI due to its large network parameters and the inclusion of a gradient penalty term for the network update.

In this study, we propose a simple and highly effective self-supervised deep learning model designed to assist the misfit function in FWI. The core concept involves transforming both observed and simulated data into a shared latent space, followed by measuring the misfit of the extracted latent features. To achieve this, we adopt a Siamese network framework (Bromley et al., 1993; Melekhov et al., 2016). The Siamese network comprises two branches: the first branch extracts features from observed data, and the second branch works on simulated data. Both branches employ an identical CNN architecture, sharing weights to apply the same filters on the data. A Euclidean distance is then employed to quantify the similarity between the features extracted from both branches. The objective of the Siamese network is to minimize the Euclidean distance loss, thereby improving the FWI performance. We test this approach on the Marmousi model and compare results with FWI supported by modern misfit functions (such as the cross-correlation loss and the Wasserstein distance) as well as with FWIGAN.

The paper is structured as follows: After this introduction, we present the theoretical development, encompassing the concept of the Siamese network and its integration into the FWI process. Subsequently, we share the application of SiameseFWI to synthetic and field examples. In the discussion section, we thoroughly explore the performance and interpretation of the proposed Siamese network. Finally, we outline the limitations of the proposed framework and provide our concluding remarks.

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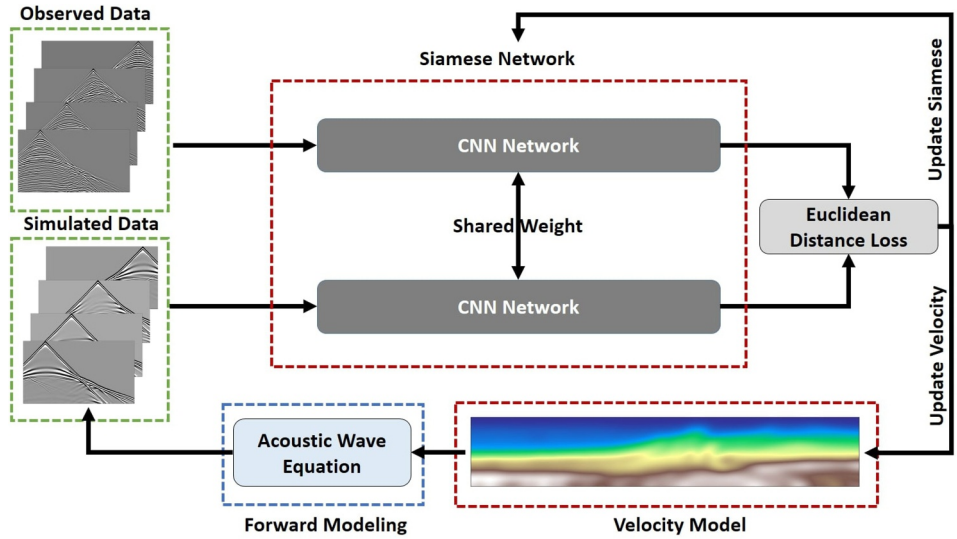


Figure 1. The framework of the proposed SiameseFWI. The SiameseFWI involves two identical Convolutional Neural Networks (CNNs) with shared weights, each taking observed and simulated data as inputs. The Euclidean distance measures the loss between the CNNs outputs, where the loss is used to optimize both the parameters of the Siamese network and the velocity model during the full-wave inversion (FWI) process. The acoustic wave equation is used for forward modeling to generate the simulated data during the FWI iteration process. The forward modeling is performed by the Deepwave toolbox, which operates using the PyTorch library.

2. Theory

The SiameseFWI comprises two key components: the forward modeling block, rooted in the acoustic wave equation, and the Siamese network responsible for extracting features from observed and simulated data to calculate the Euclidean distance loss between them. This loss is then used to update both the velocity model and the Siamese network through an iterative process. SiameseFWI employs distinct Adam optimizers for updating the velocity model and the parameters of the Siamese network separately. The refined velocity model is subsequently employed to generate simulated data. Figure 1 shows the proposed framework of SiameseFWI.

For the forward modeling, we employ a 2D acoustic wave equation with constant density expressed as follows:

$$\nabla^2 p(r,\, t) = \frac{1}{v^2(r)} \frac{\partial^2 p(r,\, t)}{\partial t^2} + s(r,t)\delta(r - r_s), \tag{1}$$

where $s(r,\, t)$ and $q(r,\, t)$ represent the pressure and the source function in terms of position (r) and time (t). r_s denotes the source location in terms of the position vector. ∇^2 and δ denote the spatial Laplace operator and the Dirac delta function, respectively. The symbol v represents the known or current (in the case of FWI) acoustic (P-wave) wave velocity, and it is a function of space, r. The aforementioned equation can be solved using a second-order finite difference in time (Smith, 1985; Virieux, 1986), where the wavefield at time $t + \Delta t$ is obtained using the wavefield from two previous time steps, that is, t and $t - \Delta t$. Accordingly, the pressure (P) can be formulated in terms of the source function (s) and the previous two-time steps of the wavefield as follows:

$$p(r,\, t + \Delta t) = v^2(r)\Delta t^2 \left[\nabla^2 p(r,\, t) - s(r,\, t)\delta(r - r_s)\right] + 2p(r,t) - p(r,\, t - \Delta t). \tag{2}$$

This allows the generation of simulated data using the acoustic wave equation based on the updated velocity model (v) for each FWI iteration. The primary purpose of the Siamese network is to extract well-represented features from both observed and simulated data, thereby enhancing the convergence of the FWI. To accomplish this, the Siamese network incorporates two identical CNN networks, sharing weights to ensure consistent feature extraction from both data.

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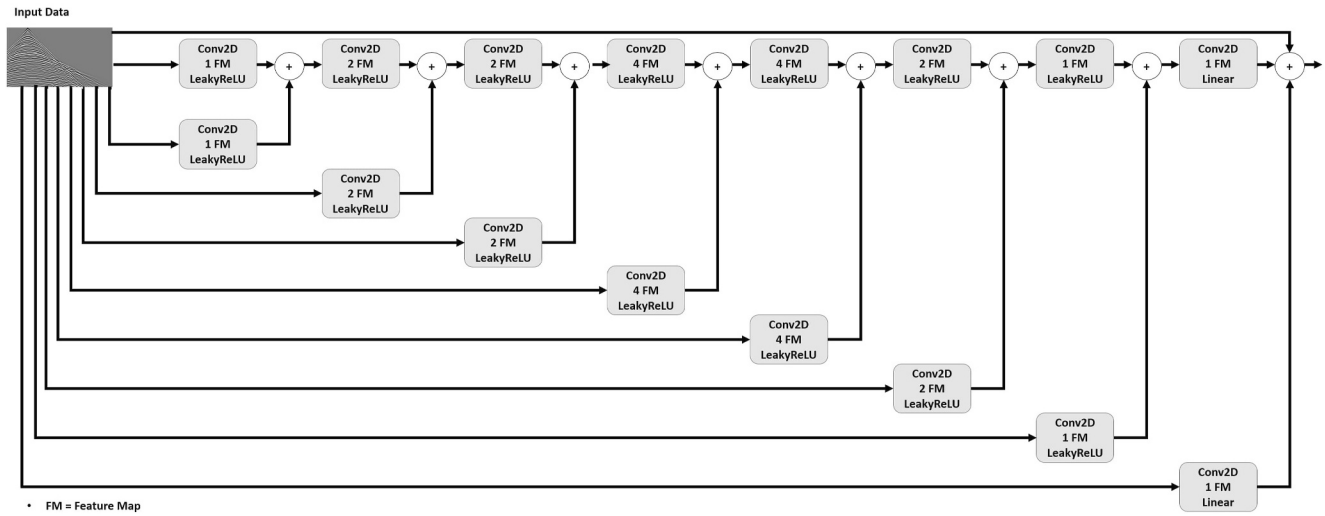


Figure 2. The architecture of the proposed convolutional neural network (CNN) network. The proposed CNN network features eight convolutional layers, each employing a LeakyReLU activation function. A skip connection is incorporated after each convolutional layer, enhancing the learning capacity of the network. The output of the CNN network represents the extracted features from either the observed or simulated data. Both branches of the Siamese network employ a similar CNN architecture.

Each CNN consists of eight 2D convolutional layers, maintaining the input size consistency without dimension reduction as shown in Figure 2. Each convolutional layer is responsible for extracting several features from the input data, with the output represented by the following expression:

$$Y_l(i,j) = \sum_{M=0}^{M-1} \sum_{N=0}^{N-1} W(m,n) X_l(i+m, j+n), \tag{3}$$

where W is the weight matrix of the convolutional layer with a kernel size of $m \times n$. X_l and Y_l denote the input and the output of the lth convolutional layer. (i, j) denotes the spatial position within the output feature map Y_l.

A kernel size ($m \times n$) of 3×3 is applied for the convolutional layers, followed by a LeakyReLU activation function with a rate of 0.1, while the last layer of the CNN uses a linear activation function. The number of feature maps is 1, 2, 2, 4, 4, 2, 1, and 1 for the eight layers from shallow to deep, respectively. To enhance the learning capacity of the CNN, eight skip connections are introduced, connecting the input and output of each convolutional layer as shown in Figure 2. These skip connections incorporate convolutional layers to further enhance feature extraction from the input data. The output of each skip connection is then added to the output of the corresponding convolutional layer. Additionally, a residual connection, in the form of an addition, connects the input data with the output of the eighth layer as shown in the upper part of Figure 2. The weights of the Siamese network are initially randomized and subsequently optimized during the FWI process. The skip connection is crucial in reconstructing input data during the initial FWI iterations, thereby preventing the prediction of random outputs and facilitating a smoother inversion process. As the FWI iterations progress, the Siamese network's weights are optimized to improve FWI inversion performance by leveraging the extracted features for comparison.

Following this, the Euclidean distance (E) is computed between the extracted features to quantify their similarity as follows:

$$E = \sqrt{\sum_{i=0}^{n} (C(d_{obs}, \phi)) - C(d_{sim}, \phi))^2}, \tag{4}$$

where d_{obs} and d_{sim} ($=p(r_s, t)$, where r_s are the locations of the receivers/sensors) are the observed and simulated data, respectively. C and ϕ are the CNN and its parameters that are optimized during the FWI process. The

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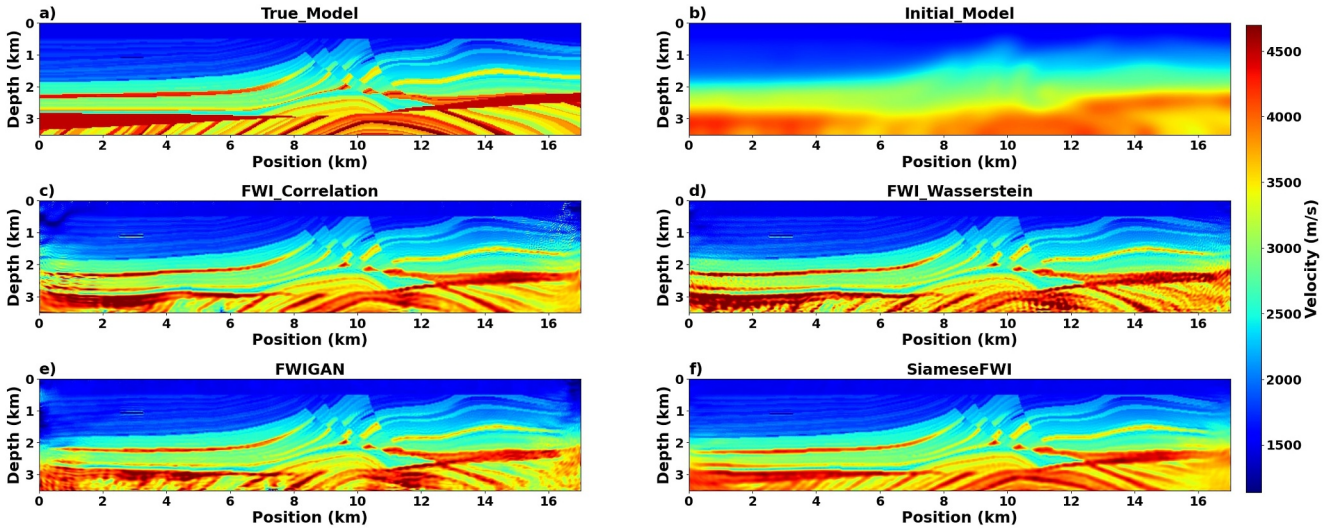


Figure 3. Results comparison for the Marmousi2 model. (a) The true Marmousi2 model. (b) The initial velocity model. The inverted model using (c) cross-correlation loss, (d) Wasserstein loss, (e) FWIGAN, and (f) SiameseFWI.

resulting Euclidean distance loss serves as the loss criterion for the FWI, guiding the inversion process toward improved convergence.

In each iteration of the FWI, we update the Siamese network parameters using Euclidean distance loss, which is based on the extracted features from both Siamese branches. Then, we update the velocity model using the updated network by propagating the loss through the Siamese network to provide us with the adjoint source. The Siamese network and velocity model are updated once per each FWI iteration. We use the Deepwave toolbox (Richardson, 2023) to conduct forward modeling, leveraging PyTorch for seamless integration with the velocity model. This integration enables a dynamic feedback loop (the lower loop of Figure 1) between the Siamese network and the velocity model, ensuring the updating of the velocity model to fit the observed seismic data better. Notably, the implementation of SiameseFWI within the PyTorch framework allows for efficient back-propagation of the gradient from the Siamese network to the velocity model. This enables the velocity model to be updated iteratively, with each update informed by the Siamese network's loss of the current data. A lower Euclidean distance value means a better match between observed and simulated data, indicating an improved inversion process for the velocity model. This emphasis contributes to improved FWI results through refined loss criteria. The proposed Siamese network is designed for simplicity and effectiveness in feature extraction. The computational cost of updating the Siamese network is relatively small compared to the modeling and solving for the adjoint wavefield (further details are provided in the Results section).

3. Results

3.1. Marmousi2 Model

To assess the efficacy of the proposed framework, SiameseFWI is employed for the inversion of the Marmousi2 model (Martin et al., 2006). The Marmousi2 model has a dimension of 117 × 567 (vertically × laterally) grid points with a spatial grid spacing of 0.03 km. The velocity values within the Marmousi2 model span a range from 1,140 to 4,700 m/s. The selected source wavelet is a Ricker wavelet characterized by a peak frequency of 5 Hz, and the total recording duration is set to 6 s. The receivers span the width of the Marmousi2 model and are placed on the surface with a uniform spacing of 0.29 km. A total of 30 sources are distributed across the surface, each separated by a horizontal interval of 0.548 km. Figure 3a shows the true model for Marmousi2. The SiameseFWI is applied to obtain the inverted model starting with the initial model shown in Figure 3b. Comparative assessment is conducted using different loss criteria, such as the cross-correlation and Wasserstein (1D Wasserstein distance) losses, resulting in the inverted models shown in Figures 3c and 3d, respectively. Additionally, the application of FWIGAN, a deep learning model, yields the inverted model shown in Figure 3e. Figure 3f shows the inverted model from SiameseFWI, demonstrating enhanced accuracy in reconstructing geological structures particularly

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Table 1
Quantitative Comparison Using Different Methods When Applied to the Marmousi2 Model

Method	SNR (*dB*)	SSIM	RMSE (*km/s*)	FWI iterations	Time consumed (*minutes*)
Cross-correlation loss	19.18	0.602	0.311	600	30
Wasserstein loss	19.42	0.588	0.303	600	33
FWIGAN	19.06	0.583	0.316	300	191
SiameseFWI	**22.57**	**0.750**	**0.211**	600	36

Note. This result is reported using one NVIDIA GPU RTX 3090. The bold numbers represent the best results.

in deeper layers when compared to other benchmark methods. For all methods, as listed in Table 1, we execute FWI for 600 iterations, other than FWIGAN for 300 iterations, as it is expensive.

To quantify the results, we calculate the signal-to-noise ratio (SNR) and structural similarity index measure (SSIM) between the true model and the inverted models obtained using various methods, as summarized in Table 1. Notably, the inverted model derived using SiameseFWI demonstrates robust performance, achieving the highest SNR and SSIM values of 22.57 dB and 0.750, respectively. The root mean square error (RMSE) between the true and inverted models is 0.311, 0.303, 0.316, and 0.211 km/s using the cross-correlation loss, Wasserstein loss, FWIGAN, and SiameseFWI, respectively.

Moreover, in Figure 4, the top row denotes shot gathers generated from various inverted velocity models, while the second row shows the differences between the generated shots from different methods and the observed shot. Notably, the SiameseFWI demonstrates a lower RMSE value of 0.14 compared to the other generated shots.

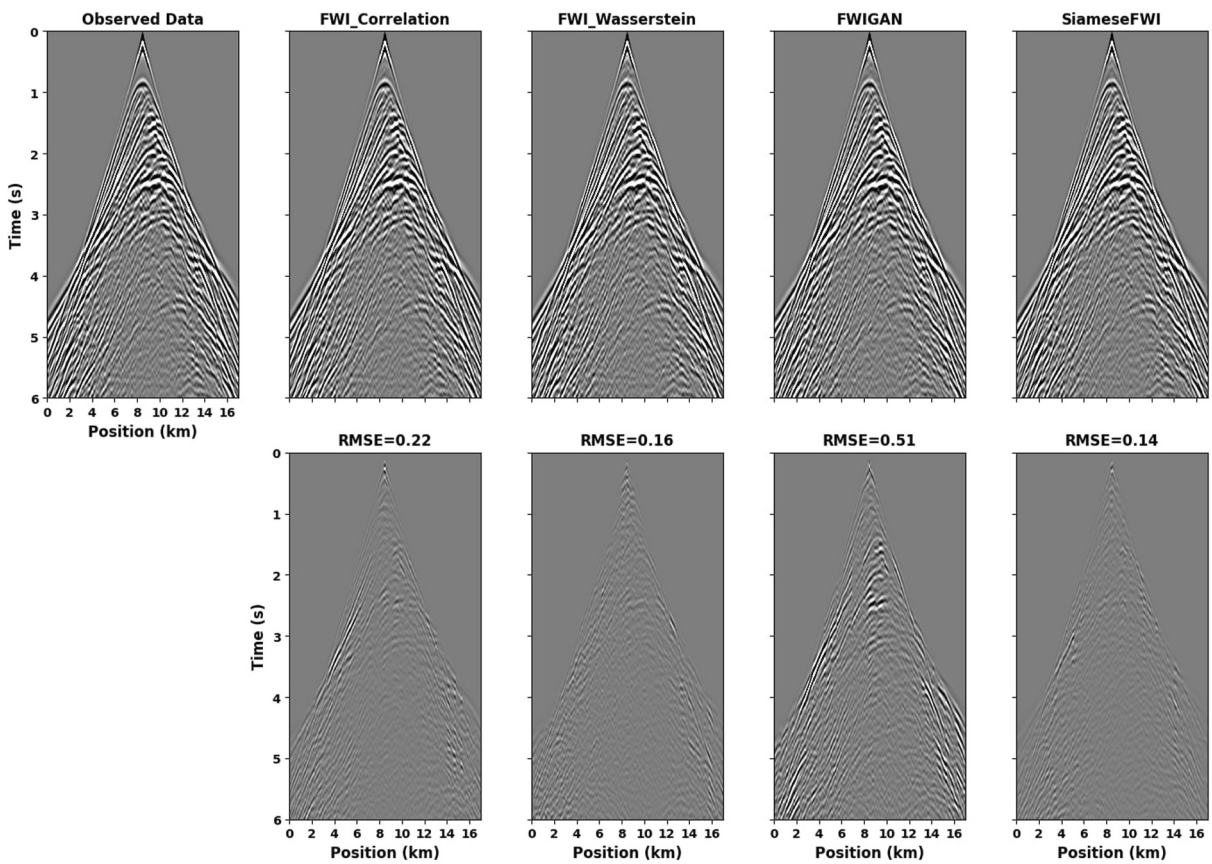


Figure 4. Top row shows shot gathers generated from different inverted velocity models compared to the true Marmousi2 velocity model. The bottom row shows the differences between the generated shots using different methods and the true Marmousi2 shot.

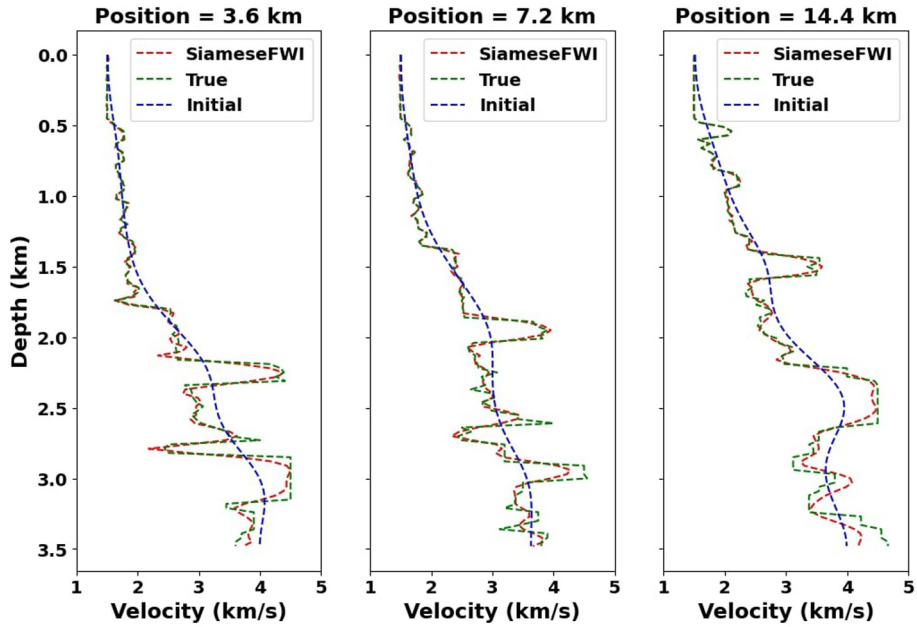


Figure 5. Velocity profiles of the Marmousi2 model for the true model, initial model, and the inverted model obtained by SiameseFWI at three locations.

Furthermore, Figure 5 shows the vertical velocity profiles at different horizontal positions of 3.6, 7.2, and 14.4 km for the true model, initial model, and inverted model obtained through SiameseFWI. The RMSE associated with the SiameseFWI for these three velocity profiles is 0.17, 0.16, and 0.17 km/s, respectively. These results indicate that the inverted model derived using SiameseFWI closely aligns with the true model, exhibiting a small margin of error. As Table 1 suggests that these superior inversion results from SiameseFWI are obtained at a reasonably small additional cost (10%–20% more) compared to the more conventional methods.

3.2. Overthrust Model

The performance of SiameseFWI is evaluated in the context of reconstructing a 2D slice extracted from the central region of the 3D Overthrust model (Aminzadeh et al., 1994). The 2D slice has dimensions of 94 × 400 with a spatial grid spacing of 0.03 km and encompasses a velocity range spanning 2,360 to 6,000 m/s. Employing a Ricker wavelet with a peak frequency of 5 Hz as the source wavelet, the sources and receivers are distributed across the surface using 30 sources with horizontal intervals of 0.387 km. The receivers span the entire lateral grid. In Figures 6a and 6b, the true and initial Overthrust velocity models are, respectively, shown. Figures 6c and 6d show the inverted models using the cross-correlation and Wasserstein loss criteria, respectively. The inverted models derived from FWIGAN and SiameseFWI are shown in Figures 6e and 6f, respectively. The SiameseFWI yields an inverted model exhibiting superior resolution with enhanced SNR when compared to the benchmark methods.

Table 2 shows the results for comparative purposes between different benchmark methods. Notably, the SiameseFWI achieves the highest SNR and SSIM at 26.98 dB and 0.725, respectively, outperforming benchmark methods. Additionally, the proposed SiameseFWI demonstrates the lowest RMSE among the benchmark methods, affirming its efficacy in accurately reconstructing the Overthrust model.

Moreover, Figure 7 shows shot gathers generated by different inverted velocity models in the first row, while the second row of Figure 7 denotes the differences between the observed data and shot gathers from various inverted models. In this instance, all generated shot gathers exhibit a small RMSE value. Notably, shot gathers generated using the inverted models employing Wasserstein loss and SiameseFWI demonstrate the lowest RMSE value of 0.02.

Furthermore, the vertical velocity profiles at distinct horizontal positions of 1.8, 3.6, and 7.2 km are presented in Figure 8, showing the true, initial, and inverted models obtained through SiameseFWI. The RMSE values for the

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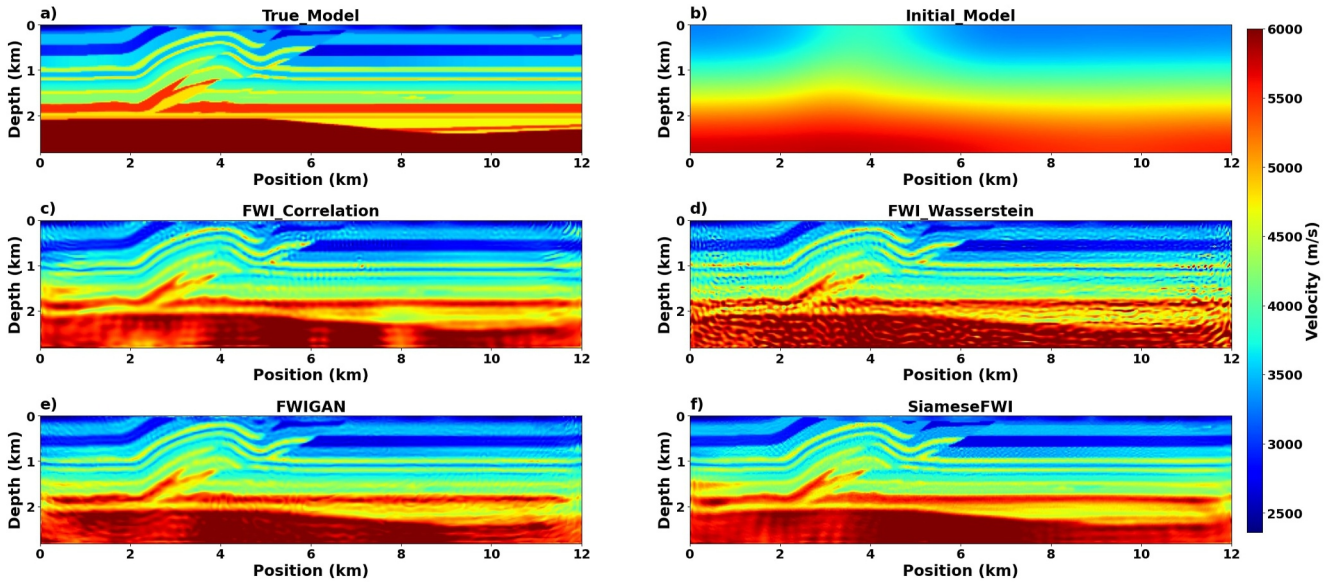


Figure 6. Results comparison for the Overthrust model. (a) The true Overthrust model. (b) The initial velocity model. The inverted model using (c) cross-correlation loss, (d) Wasserstein loss, (e) FWIGAN, and (f) SiameseFWI.

three velocity profiles, when compared to the true model, are 0.17, 0.18, and 0.11 km/s. Notably, the vertical velocity profiles show that the inverted model using SiameseFWI aligns remarkably well with the true Overthrust velocity model.

Besides the robust performance of the proposed SiameseFWI, it is again imperative to highlight that SiameseFWI introduces minimal computational overhead to the traditional FWI, requiring only about 10%–20% of additional cost for the accurate inversion of the Marmousi2 and Overthrust models. In contrast, FWIGAN, a deep learning model, seriously extends the inversion time by sixfold while yielding comparable FWI performance.

3.3. Field Example

Finally, we evaluate the performance of the proposed SiameseFWI using real field data obtained from marine seismic surveys in the northwestern part of Australia. The data consist of 116 airgun-ignited shot records, spanning a lateral extent of approximately 20 km, with shots spaced around 90 m apart. Each shot is recorded for 7 s, employing a streamer equipped with 648 hydrophones spaced at 12.5 m intervals, resulting in a total offset of approximately 8.1 km.

The source wavelet is determined by estimating the near-offset recording of the direct arrival. Source inversion is specifically applied to the direct arrival portion of the data, utilizing a model with a water velocity of 1.5 km/s during inversion. Spectrum analysis reveals that the lowest available frequency is approximately 3 Hz. Several pre-processing steps are implemented, including bandpass filtering and muting of pre-arrival noise. To reduce the inversion cost considering the frequency range we are addressing here, the data are resampled in the spatial axis from 12.5 to 25 m and in the temporal axis from 1 to 2 ms.

Table 2
Quantitative Comparison Using Different Methods When Applied to the Overthrust Model

Method	SNR (*dB*)	SSIM	RMSE (*km/s*)	FWI iterations	Time consumed (*minutes*)
Cross-correlation loss	24.52	0.663	0.218	600	29
Wasserstein loss	23.25	0.547	0.315	600	32
FWIGAN	25.95	0.663	0.228	300	167
SiameseFWI	**26.98**	**0.725**	**0.211**	600	34

Note. This result is reported using one NVIDIA GPU RTX 3090. The bold numbers represent the best results.

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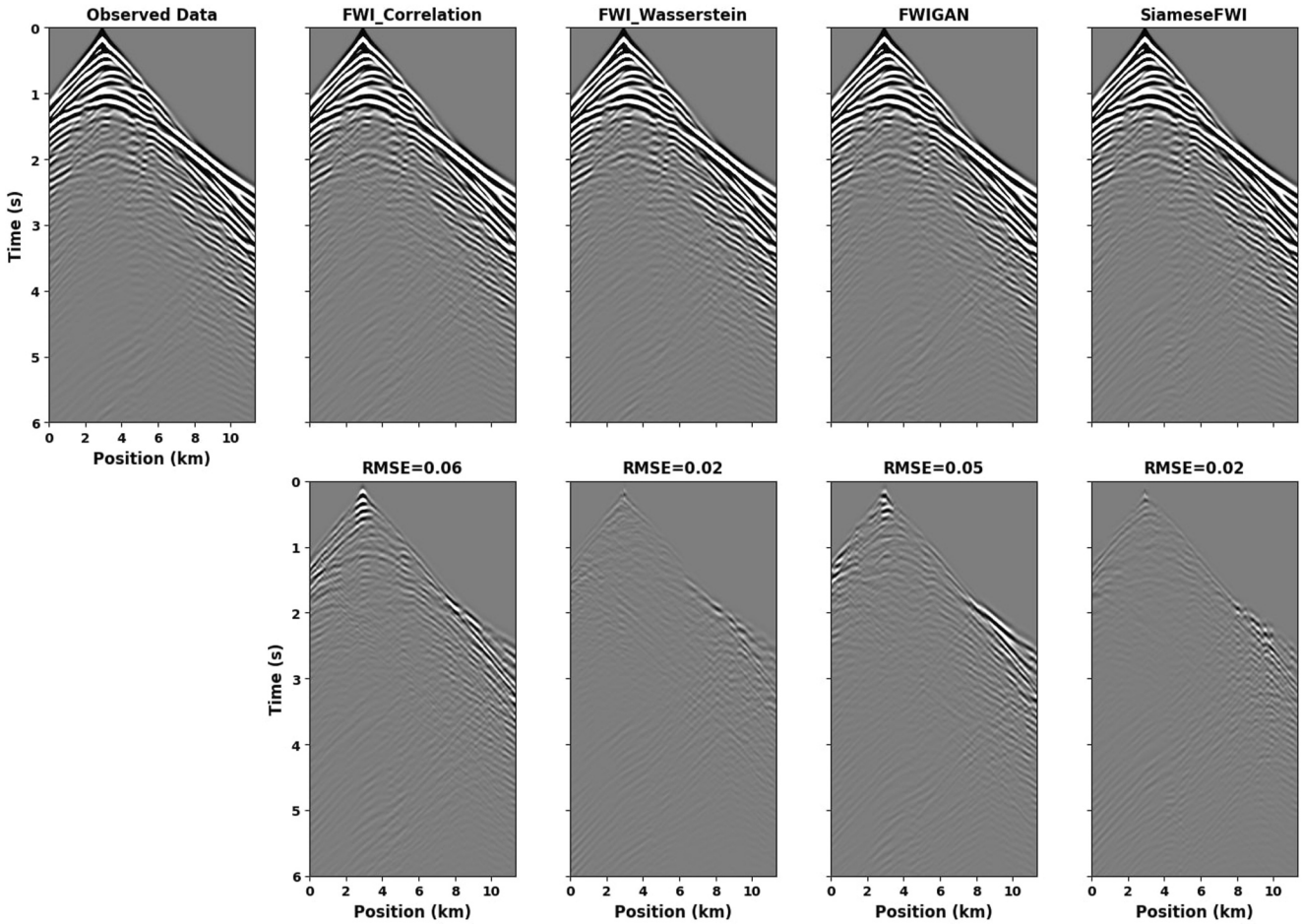


Figure 7. Top row shows shot gathers generated from different inverted velocity models compared to the true Overthrust velocity model. The bottom row shows the differences between the generated shots using different methods and the true Overthrust shot.

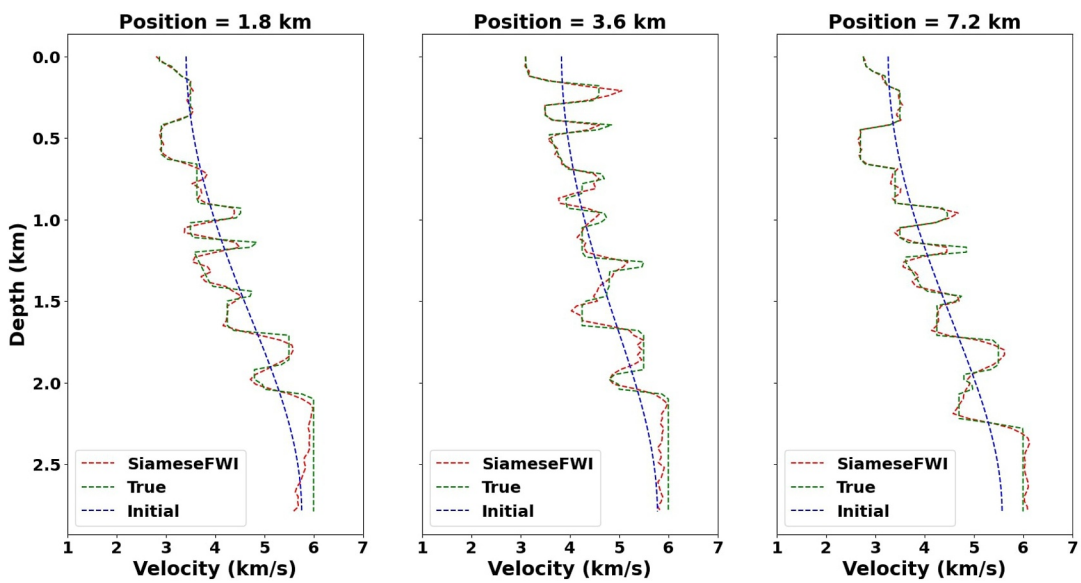


Figure 8. Velocity profiles of the Overthrust model for the true model, initial model, and the inverted model obtained by SiameseFWI at three locations.

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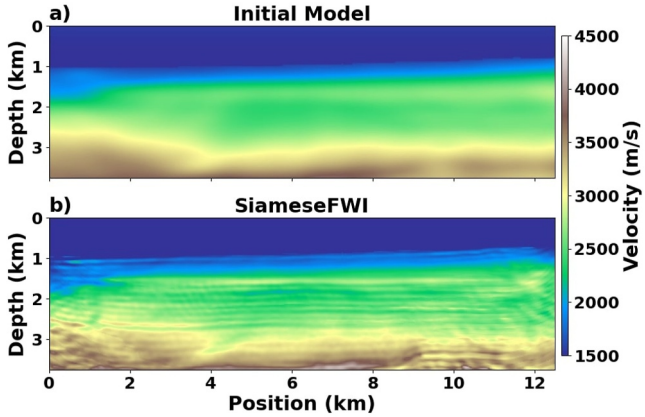


Figure 9. Application of SiameseFWI to the data acquired by CGG in the North Western part of Australia. (a) The initial velocity model. (b) The inverted model using SiameseFWI.

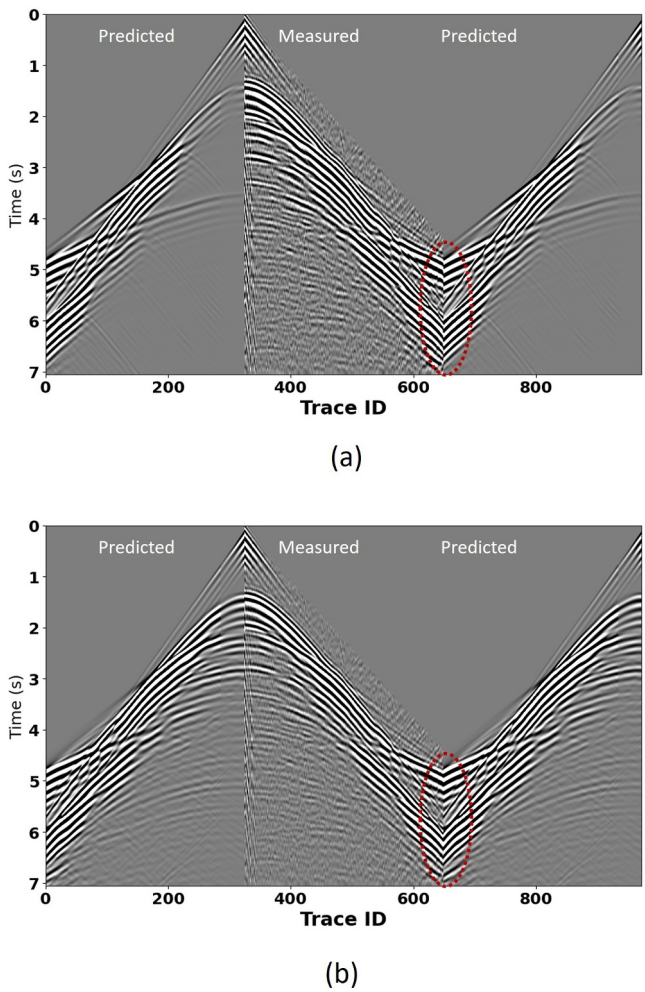


Figure 10. Example of the shot gathers generated by (a) the initial model and (b) the inverted model obtained by SiameseFWI.

After preprocessing, the filtered data below <10 Hz are used as the observed data for SiameseFWI. The initial velocity model, shown in Figure 9a, is derived from the migration velocity analysis results (Kalita & Alkhalifah, 2017). The model grid has a spacing of 25 m both laterally and vertically. During FWI iterations, SiameseFWI optimizes its parameters to extract features from each Siamese branch. The first branch takes the observed data as input, while the second branch uses the simulated data. Both branches share a CNN architecture with shared weights, applying identical convolutional kernels at various levels for simulated and observed data. The Adam optimizer updates the velocity model with a step length (learning rate) set to 10, and the Siamese network uses a learning rate of 1e−4. The process involves 200 FWI iterations, aiming to minimize the loss between Siamese outputs from the two branches. Figure 9b shows the inverted velocity model obtained through SiameseFWI.

For an in-depth assessment of the reliability of the inverted model, a comprehensive comparison is conducted, contrasting the generated shot gathers using both the initial and inverted velocity models, as shown in Figure 10. Notably, the shot gather derived from the inverted model exhibits better data fitting to the observed data than the initial (migration obtained) model. This improvement is particularly noticeable for far offsets, as highlighted by the red ellipses in Figure 10. This observation emphasizes the effectiveness of the inversion process in capturing subsurface characteristics.

We performed reverse time migration for the initial and inverted models to obtain their corresponding images as shown in Figure 11. A notable improvement can be observed at depths 1.5–2 km, where the interfaces are better resolved in the inverted model (indicated by red arrows in Figure 11a) compared to that of the initial model (red arrows in Figure 11b). For a better comparison, we calculated the angle gather at position 1–11 km (with 1 km spacing) of the initial and inverted models as shown in Figures 12a and 12b, respectively. While the angle gathers of both velocity models are mostly flat, we noticed that the energy is more focused at depths 1.5–2 km in the inverted model (indicated by red ellipses in Figure 12), proving the value of our inversion process.

4. Discussion

4.1. Interpretation of the Siamese Network

To interpret the performance of SiameseFWI, we visualize the output of the Siamese network, as shown in Figure 13. Figures 13a and 13c show shot gathers generated by the initial and true Overthrust model, respectively. Correspondingly, Figures 13c and 13d show the output of the Siamese network. Notably, the Siamese network highlights the earlier P-wave arrival times, especially at the far offsets, of both observed and simulated data, facilitating a more effective comparison and focusing on more transmission components. This analysis reveals that rather than comparing entire shot gathers, including possible reflections in the observed data, the Siamese network learns to focus on the crucial events for the inversion, such as events necessary to update the background first. Thus, if the highlighted P-waves from the observed data, extracted by the Siamese network, match those extracted from the simulated data, it indicates that the FWI process is converging. Since both branches of the Siamese network share the same weights, there are two options for the network: either to make the observed data resemble the simulated data or to enhance the simulated data to closely

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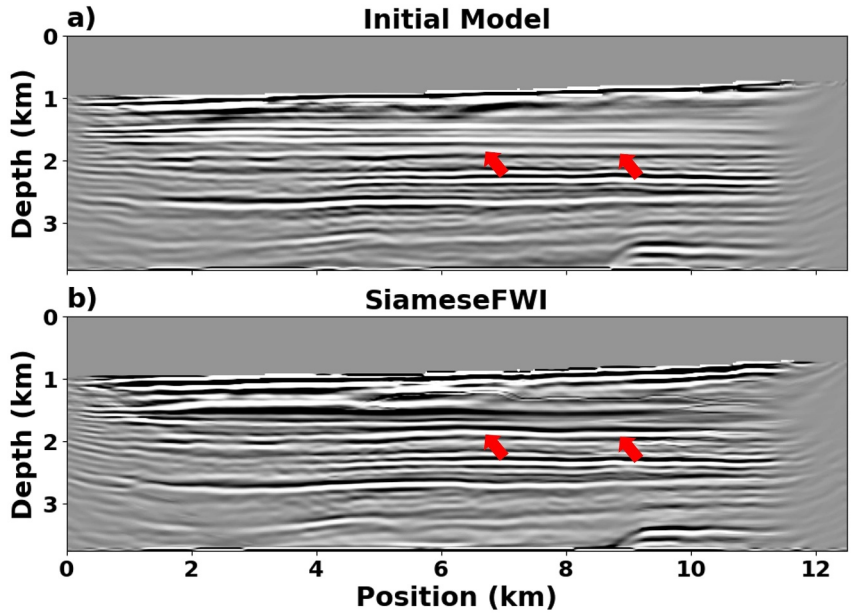


Figure 11. RTM images using (a) the initial model and (b) the SiameseFWI inverted model.

match the observed data for improved comparison. As shown in Figure 13, the network found it easier to extract the important features, particularly those related to the P-wave arrival times, from the simulated data and compare them to the observed data, resulting in a robust FWI inversion process.

In further analysis of the network's performance, we employ the SiameseFWI framework without weight sharing using different kernels for simulated and observed data. The resulting inverted model achieves an SNR and SSIM of 24.37 dB and 0.64. The output of the Siamese network, as shown in Figure 14, highlights features associated with P-wave arrival times in both simulated and observed data. This consistent emphasis highlights the importance of these features in ensuring a robust inversion process within the SiameseFWI framework.

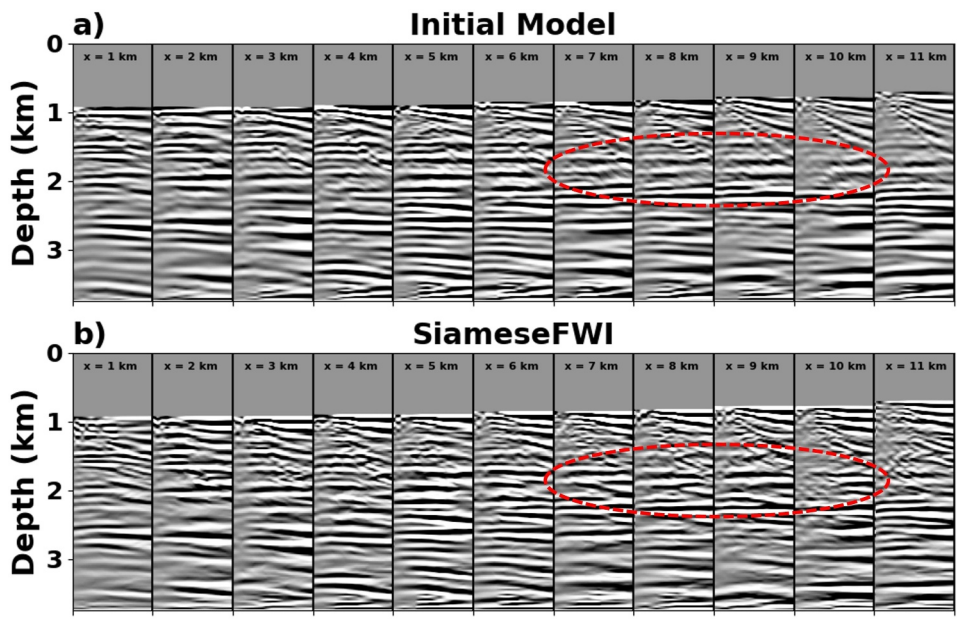


Figure 12. Angle gathers at position 1–11 km with an increment of 1 km using the initial model (a) and the SiamseFWI inverted model (b). Each angle gather spans up to 60°.

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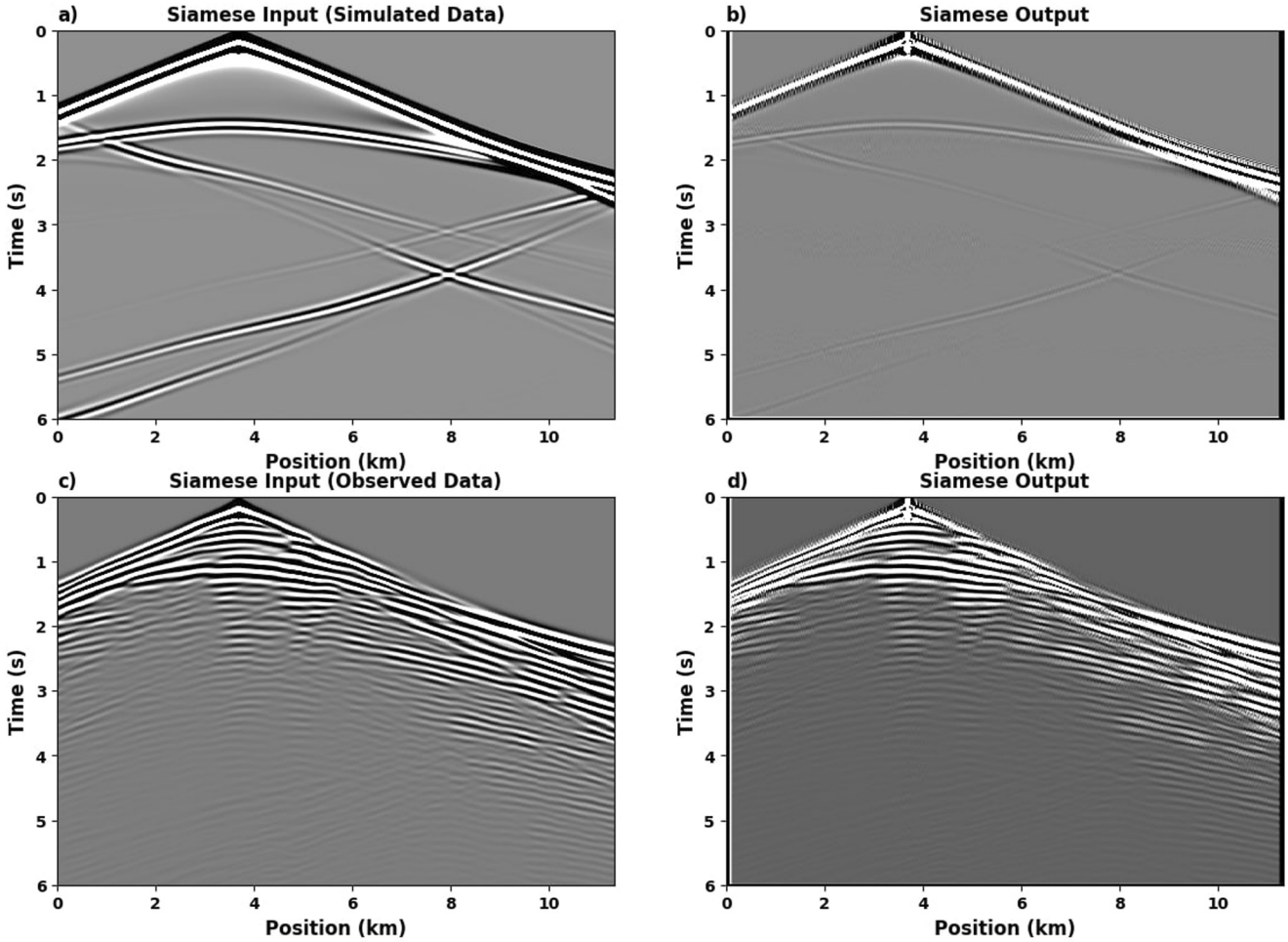


Figure 13. Interpretation of the Siamese network. Shot gathers generated using (a) the initial Overthrust model and (c) the true Overthrust model. (b) The output of the Siamese network for input (a). (d) The output of the Siamese network for input (c).

4.2. The Sensitivity of the SiameseFWI to the Noise

In evaluating the robustness of SiameseFWI under noisy conditions, we introduce additive white Gaussian noise (AWGN) to the simulated data, assessing the network's capacity to effectively invert the Overthrust model. This examination aims to demonstrate whether the Siamese network can learn to extract the significant features within the noisy simulated data (such as denoising) irrespective of the added noise. AWGN is incorporated into the simulated data, resulting in an SNR of −5 dB.

Figures 15a and 15b show the inverted models obtained using Wasserstein loss and SiameseFWI, respectively. The SNR values for the inverted models from the SiameseFWI using Wasserstein loss and SiameseFWI are 19.18 and 23.48 dB, while the SSIM scores are 0.39 and 0.65, respectively. Remarkably, the SiameseFWI-inverted model exhibits a better inverted result, emphasizing its effectiveness in mitigating the impact of noise on the inversion process.

4.3. The Sensitivity of the SiameseFWI to the Initial Model

To further evaluate the efficacy of SiameseFWI, we use a linear initial model, as shown in Figure 16b, to invert the Overthrust velocity model. The Adam optimizer is employed with a scheduled learning rate to iteratively update the parameters of the Siamese network. Initially, a small learning rate of 1e−6 is set, subsequently increased by a factor of 10 every 200 FWI iterations. This gradual adjustment of the learning rate ensures a smooth parameter

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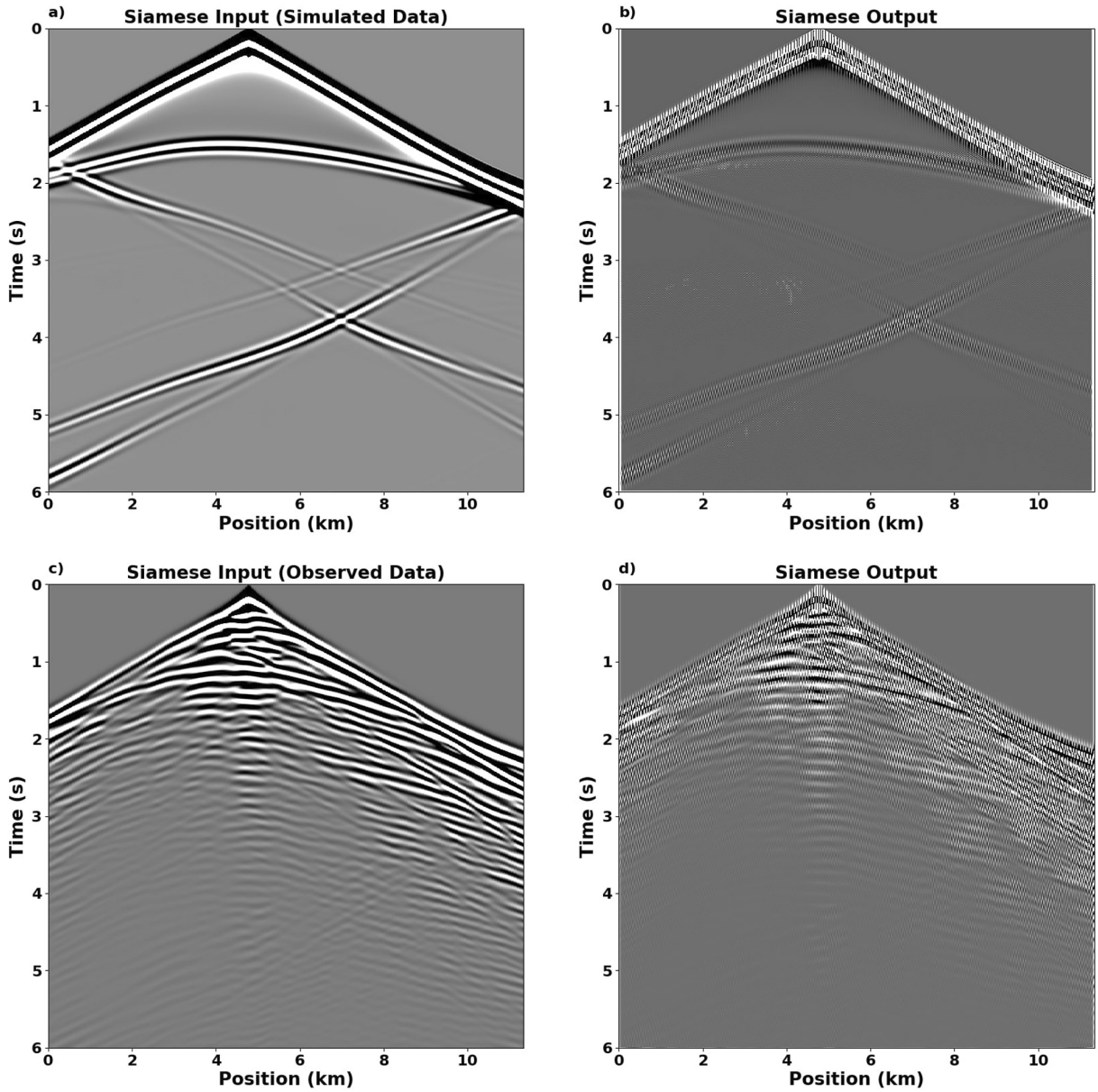


Figure 14. Interpretation of the Siamese network using different weights for its branches. Shot gathers are generated using (a) the initial Overthrust model and (c) the true Overthrust model. (b) The output of the Siamese network for input (a). (d) The output of the Siamese network for input (c).

update for the Siamese network, enabling the secondary Adam optimizer associated with the velocity model to progressively reconstruct the Overthrust model's structure.

After 400 FWI iterations, the learning rate for the Siamese network parameters is increased to 1e−4, facilitating further refinement of the inverted model. This adaptive learning rate strategy effectively mitigates the risk of local minima and yields an accurate inversion of the Overthrust model, as shown in Figure 16d. The SNR and SSIM metrics for the SiameseFWI-inverted model reach 23.39 dB and 0.68, respectively. In comparison, the model inverted using the Wasserstein loss exhibits an SNR of 15.68 dB and SSIM of 0.53. The RMSE values for the inverted models using Wasserstein loss and SiameseFWI are 0.27 and 0.72, respectively.

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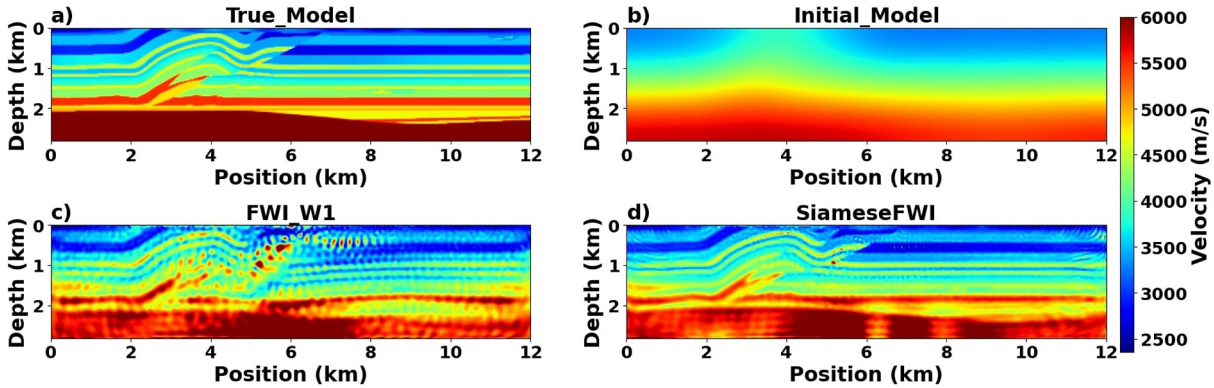


Figure 15. Performance of the SiameseFWI when adding noise to the simulated data. (a) The true Overthrust model. (b) The initial model. The inverted model using (c) Wasserstein loss and (d) SiameseFWI.

4.4. The Flexibility of the Loss Function

Siamese network's efficacy lies in its ability to extract meaningful features from both observed and simulated data, facilitating improved convergence in FWI. In addition to employing the Euclidean distance as a loss function, alternative loss functions such as Wasserstein and cross-correlation are explored for the SiameseFWI. Figure 17 shows the inverted models generated using different loss functions, revealing high-resolution and clean inverted models in comparison to the true Overthrust model.

SNRs for the inverted models, using Euclidean, Wasserstein, cross-correlation, and L_2 losses, are 26.98, 27.74, 25.24, and 25.59 dB, respectively. Furthermore, the SSIM for the inverted models using the four different loss functions is computed as 0.725, 0.675, 0.674, and 0.667, demonstrating the quality of the inverted models.

This analysis emphasizes the versatility of applying different loss functions to the Siamese network's output, thereby enhancing the FWI process. It also highlights the Siamese network's ability to extract significant features from both observed and simulated data, making it adaptable to various loss functions and further enhancing the FWI process.

4.5. The Sensitivity of the SiameseFWI to the Waveform Amplitude and an Unknown Source Wavelet

To assess the influence of waveform amplitude on SiameseFWI, we normalized both the simulated and observed data from the Overthrust model and performed the SiameseFWI framework. However, this normalization resulted in an estimated loss that became excessively small, hindering the updating of the Siamese network weights. To address this, we increased the learning rate of the Adam optimizer associated with the velocity model. The inverted velocity model yielded an SNR of 26.01 dB and SSIM of 0.67. Figure 18c shows the inverted velocity

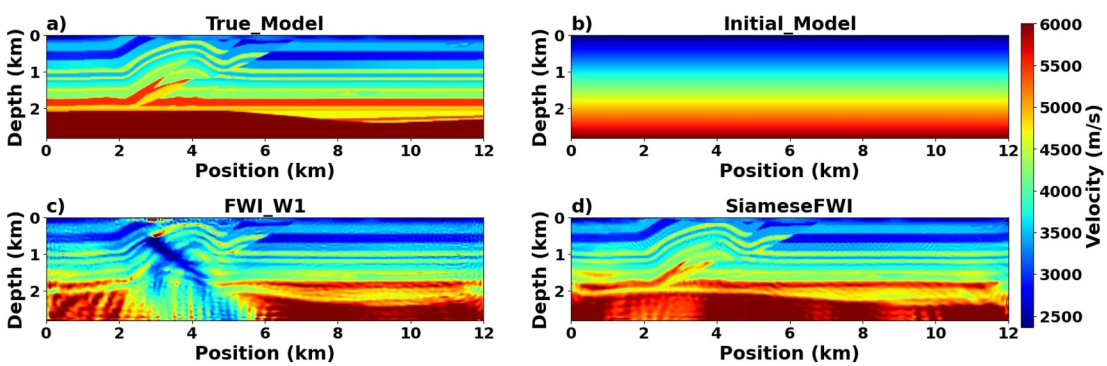


Figure 16. Performance of the SiameseFWI using a linear initial model. (a) The true Overthrust model. (b) The linear initial model. The inverted model using (c) Wasserstein loss and (d) SiameseFWI.

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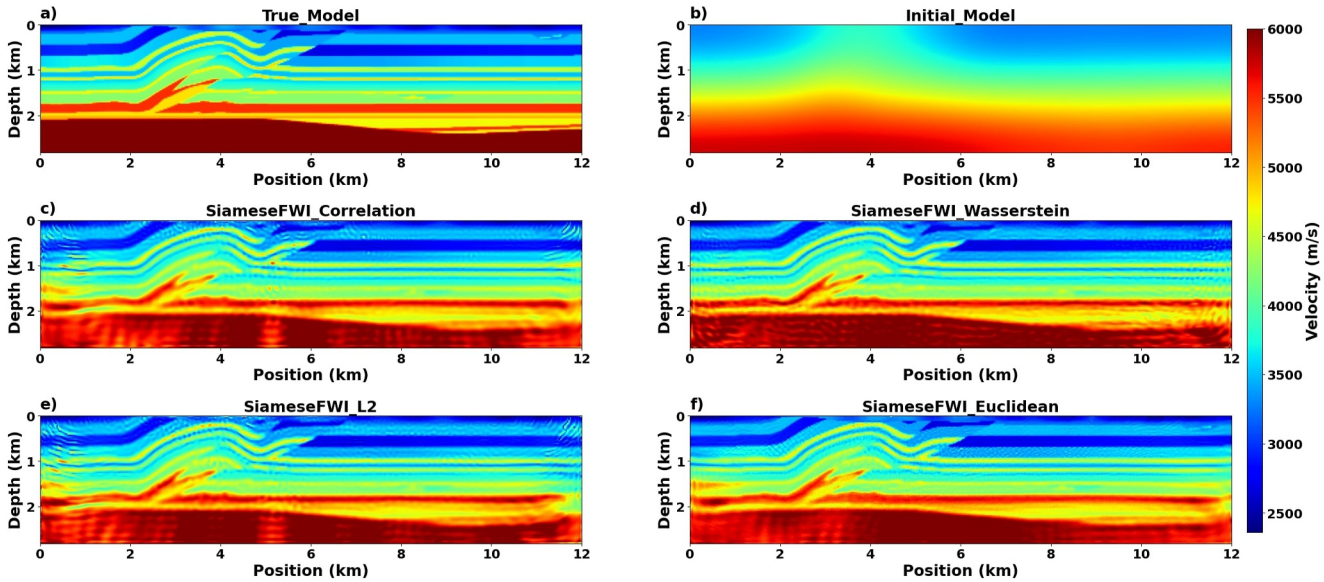


Figure 17. Performance of the SiameseFWI using different loss functions. (a) The true Overthrust model. (b) The initial model. The inverted model was obtained by SiameseFWI using (c) cross-correlation loss, (d) Wasserstein loss, (e) L_2 loss, and (f) Euclidean loss.

model. Nonetheless, the SiameseFWI works better without normalizing the waveform due to the resulting small loss values, which prevents efficient updating of the Siamese network.

Furthermore, we evaluate the SiameseFWI in scenarios where the source wavelet is unknown. First, we initialized the source wavelet with an incorrect peak frequency of 10 Hz. Subsequently, we introduced a new optimizer to update the source wavelet during the FWI process. Using the estimated loss from the Siamese network, we updated not only the Siamese network and the velocity model but also the source wavelet. Despite these complexities, the resulting inverted model achieved promising results, with an SNR of 24 dB and an SSIM of 0.65. Figure 18d shows the inverted velocity model. This demonstrates the robustness of the SiameseFWI in inverting for the velocity and source wavelet simultaneously.

4.6. SiameseFWI and Transfer Learning

We evaluated the adaptability of the Siamese network to different data after training. Initially, we used pre-trained weights from the Overthrust model to initialize the Siamese network for inverting the Marmousi2 model. Subsequently, we fine-tuned the Siamese network weights during the FWI process. The resulting inverted model achieved an SNR of 20.27 dB and an SSIM of 0.67, indicating a decrease in performance compared to the original framework. Therefore, it is preferable to optimize the Siamese network for each new data during the FWI process

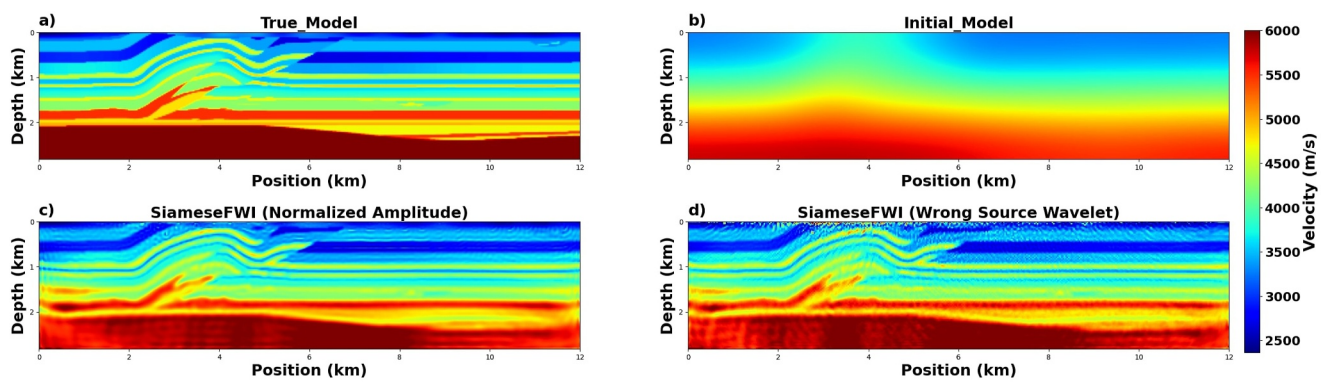


Figure 18. Performance of the SiameseFWI using normalized waveform and wrong source wavelet. (a) The true Overthrust model. (b) The initial model. The inverted model was obtained by SiameseFWI using (c) the normalized waveform of both simulated and observed data and (d) wrong source wavelet.

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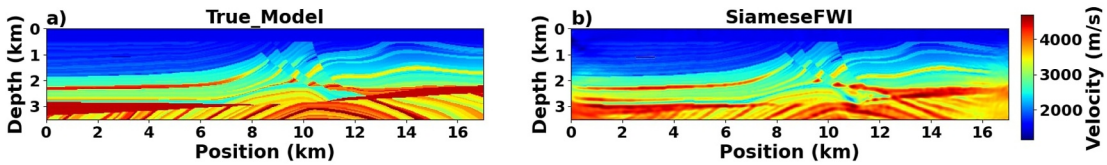


Figure 19. Effect of transfer learning within the SiameseFWI framework. (a) True Marmousi2 model. (b) The inverted model was obtained by the SiameseFWI framework and transfer learning.

to ensure optimal performance. The reason for that is the SiameseFWI starts the inversion by mainly comparing the observed and simulated data through the residual connection, and thus, providing updates similar to conventional FWI. The Siamese weights learned to optimize the inversion for another data set will alter this feature. In addition, these weights are dynamic as they evolve with the evolution of the velocity model. So the Siamese weights learned at the end of the FWI process are probably optimal for the final iterations as the predicted velocity model becomes close to the true one. Figure 19b shows the inverted model using the transfer learning strategy.

4.7. The Limitation of SiameseFWI and the Future Work

Given the concurrent use of two Adam optimizers for updating both the Siamese network and the velocity model, ensuring proper coordination between the learning rates of the two Adam optimizers is crucial for a successful inversion. A smaller learning rate for the optimizer responsible for updating the Siamese network allows sufficient time for the second optimizer to update the velocity model. This strategy helps prevent the Siamese network from updating too quickly and potentially overshadowing the updates to the velocity model, thus facilitating a more effective and balanced optimization process.

Since the architecture of the proposed Siamese network did not include a pooling layer, the size of the input data remains constant across all Siamese layers. While this can be advantageous for preserving spatial information, it may indeed lead to challenges when processing field data with large dimensions. One potential issue is the increased computational cost associated with processing large-dimensional field data. Without pooling layers to downsample the input data, the network may require more memory and processing power to handle the larger input size. This could result in longer training times and higher resource requirements.

Further investigations are needed to figure out how to further guide the Siamese network for improved feature extraction, enhancing the comparison between simulated and observed data and possibly imposing some regularization on the Siamese network. This may imply in the future that the Siamese network learns also to attenuate S-waves in observed field data when we use the acoustic assumption for modeling.

5. Conclusions

Our proposed enhancement, SiameseFWI, for the FWI process involves integrating a Siamese network into FWI. This network transforms observed and simulated data into a latent space by learning significant features to minimize the FWI loss and update the velocity model iteratively. The Euclidean distance is used to measure the loss between the extracted features from both observed and simulated data. Comprising two CNNs with shared weights, the Siamese network ensures the extraction of consistent features from both observed and simulated data. This incorporation results in improved FWI performance without imposing a significant computational cost, distinguishing it from other benchmark self-supervised deep learning models. Moreover, SiameseFWI demonstrates robust inversion when applied to field data from the western part of Australia, highlighting its versatility and generalization features. Choosing the optimal loss function for SiameseFWI could be one of the interesting points that can further enhance the inversion process. Moreover, additional investigations are needed to enhance the feature extraction capabilities of the Siamese network, thereby improving the comparison between observed and simulated data and consequently improving the FWI convergence process.

Data Availability Statement

The data and accompanying codes that support the findings of this study are available (Saad, 2024).

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